

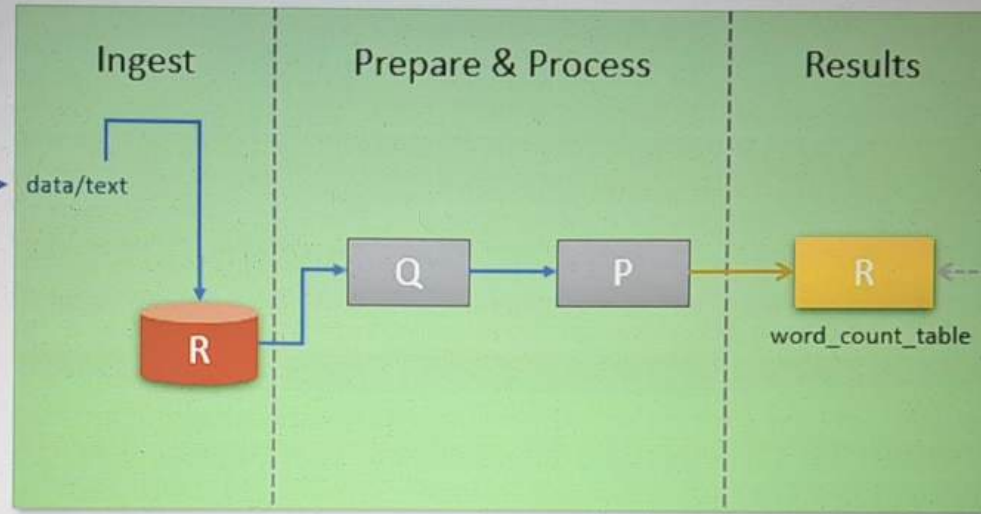
Source System

Data File 1

Data File 2

Data File 3

Data Engineering Platform



Consumer Application

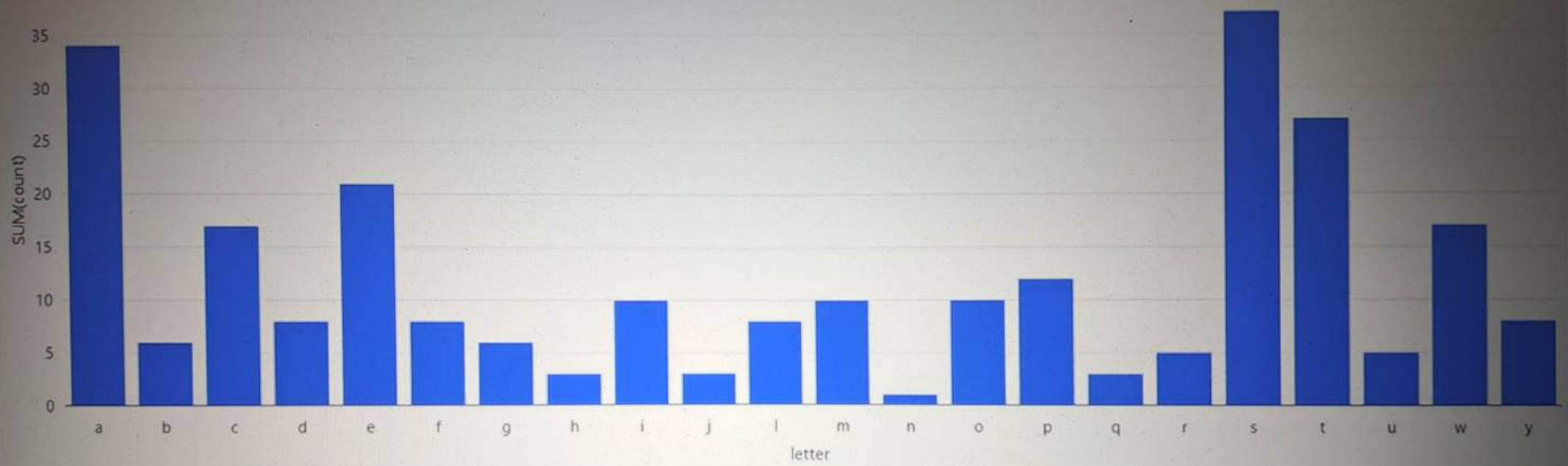


Word By Letters

What is the frequency?

15 minutes





02-word-count-ingestion

Python ▾

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Cmd 1

#Cleanup and Setup

```
base_data_dir = "/FileStore/data_spark_streaming_scholarnest"
```

```
spark.sql("drop table if exists word_count_table")
```

```
dbutils.fs.rm("/user/hive/warehouse/word_count_table", True)
```

```
dbutils.fs.rm(f"{base_data_dir}/chekpoint", True)
```

```
dbutils.fs.rm(f"{base_data_dir}/data/text", True)
```

```
dbutils.fs.mkdirs(f"{base_data_dir}/data/text")
```

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01-streaming-word-count Python

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Cmd 1

```
base_data_dir = "/FileStore/data_spark_streaming_scholarnest"
```

Python



Cmd 2

```
lines = (spark.read
        .format("text")
        .option("lineSep", ".")
        .load(f"{base_data_dir}/data/text"))
```

Python



Cmd 3

```
from pyspark.sql.functions import explode, split, trim, lower

raw_words = lines.select(explode(split(lines.value, " ")).alias("word"))

quality_words = (raw_words.select(lower(trim(raw_words.word)).alias("word"))
                  .where("word is not null"))
```



01-streaming-word-count

Python ▾

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Cmd 3

```
from pyspark.sql.functions import explode, split, trim, lower

raw_words = lines.select(explode(split(lines.value, " ")).alias("word"))

quality_words = ( raw_words.select(lower(trim(raw_words.word)).alias("word"))
                  .where("word is not null")
                  .where("word rlike '[a-z]')")

wordCounts = quality_words.groupBy("word").count()
```

- ▶ raw_words: pyspark.sql.dataframe.DataFrame = [word: string]
- ▶ quality_words: pyspark.sql.dataframe.DataFrame = [word: string]
- ▶ wordCounts: pyspark.sql.dataframe.DataFrame = [word: string, count: long]

Command took 0.77 seconds -- by prashantkumarpandey@gmail.com at 10/7/2023, 6:19:27 PM on my

Cmd 4



Udemy

01-streaming-word-count

Python ▾

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```
raw_words: pyspark.sql.dataframe.DataFrame = [word: string]
quality_words: pyspark.sql.dataframe.DataFrame = [word: string]
wordCounts: pyspark.sql.dataframe.DataFrame = [word: string, count: long]
```

Command took 0.77 seconds -- by prashantkumarpandey@gmail.com at 10/7/2023, 6:19:27 PM on my cluster

Cmd 4

```
(wordCounts.write
    .format("delta")
    .mode("overwrite")
    .saveAsTable("word_count_table")
)
```

Python

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```

class batchWC():
    def __init__(self):
        self.base_data_dir = "/FileStore/data_spark_streaming_scholarnest"

    def getRawData(self):
        from pyspark.sql.functions import explode, split
        lines = (spark.read
                  .format("text")
                  .option("lineSep", ".")
                  .load(f"{self.base_data_dir}/data/text")
                  )
        return lines.select(explode(split(lines.value, " ")).alias("word"))

    def getQualityData(self, rawDF):
        from pyspark.sql.functions import trim, lower
        return ( rawDF.select(lower(trim(rawDF.word)).alias("word"))
                  .where("word is not null")
                  .where("word rlike '[a-z]')
                  )

    def getWordCount(self, qualityDF):
        return qualityDF.groupBy("word").count()

    def overwriteWordCount(self, wordCountDF):

```

```
from pyspark.sql.functions import trim, lower
return ( rawDF.select(lower(trim(rawDF.word)).alias("word"))
        .where("word is not null")
        .where("word rlike '[a-z]')
        )
```

```
def getWordCount(self, qualityDF):
    return qualityDF.groupBy("word").count()
```

```
def overwriteWordCount(self, wordCountDF):
    ( wordCountDF.write
      .format("delta")
      .mode("overwrite")
      .saveAsTable("word_count_table")
    )
```

```
def wordCount(self):
    print(f"\tExecuting Word Count...", end='')
    rawDF = self.getRawData()
    qualityDF = self.getQualityData(rawDF)
    resultDF = self.getWordCount(qualityDF)
    self.overwriteWordCount(resultDF)
    print("Done")
```



Cmd 1

```
%run ./01-streaming-word-count
```



Cmd 2

```
class batchWCTestSuite(self):
    def __init__(self):
        self.base_data_dir = "/FileStore/data_spark_streaming_scholarnest"

    def cleanTests(self):
        print(f"Starting Cleanup...", end='')
        spark.sql("drop table if exists word_count_table")
        dbutils.fs.rm("/user/hive/warehouse/word_count_table", True)

        dbutils.fs.rm(f"{base_data_dir}/checkpoint", True)
        dbutils.fs.rm(f"{base_data_dir}/data/text", True)

        dbutils.fs.mkdirs(f"{base_data_dir}/data/text")
        print("Done\n")
```

Python



02-word-count-test-suite

Python ▾

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```
print("Done\n")

def ingestData(self, itr):
    print(f"\tStarting Ingestion...", end='')
    dbutils.fs.cp(f"{self.base_data_dir}/datasets/text/text_data_{itr}.txt", f"{self.
    base_data_dir}/data/text/")
    print("Done")

def assertResult(self, expected_count):
    actual_count = spark.sql("select sum(count) from word_count_table where substr(word,
    1, 1) == 's'").collect()[0][0]
    assert expected_count == actual_count, f"Test failed! actual count is {actual_count}"

def runTests(self):
    self.cleanTests()
    wc = batchWC()

    print("Testing first iteration of batch word count...")
    self.ingestData(1)
    wc.wordCount()
    self.assertResult(25)
    print("First iteration of batch word count completed.\n")
```



02-word-count-test-suite Python ▾

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▶ Run all

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```
wc = batchWC()

print("Testing first iteration of batch word count...")
self.ingestData(1)
wc.wordCount()
self.assertResult(25)
print("First iteration of batch word count completed.\n")

print("Testing second iteration of batch word count...")
self.ingestData(2)
wc.wordCount()
self.assertResult(32)
print("Second iteration of batch word count completed.\n")

print("Testing third iteration of batch word count...")
self.ingestData(3)
wc.wordCount()
self.assertResult(37)
print("First iteration of batch word count completed.\n")
```



02-word-count-test-suite

Python ▾

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Interrupt

my cluster ▾

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```
wc.wordCount()  
self.assertResult(37)  
print("First iteration of batch word count completed.\n")
```

•• Waiting to run...

Cmd 3

```
bwcTS = batchWCTestSuite()  
bwcTS.runTests()
```

•• Waiting to run...

Shift+Enter to run

Shift+Ctrl+Enter to run selected text

Python

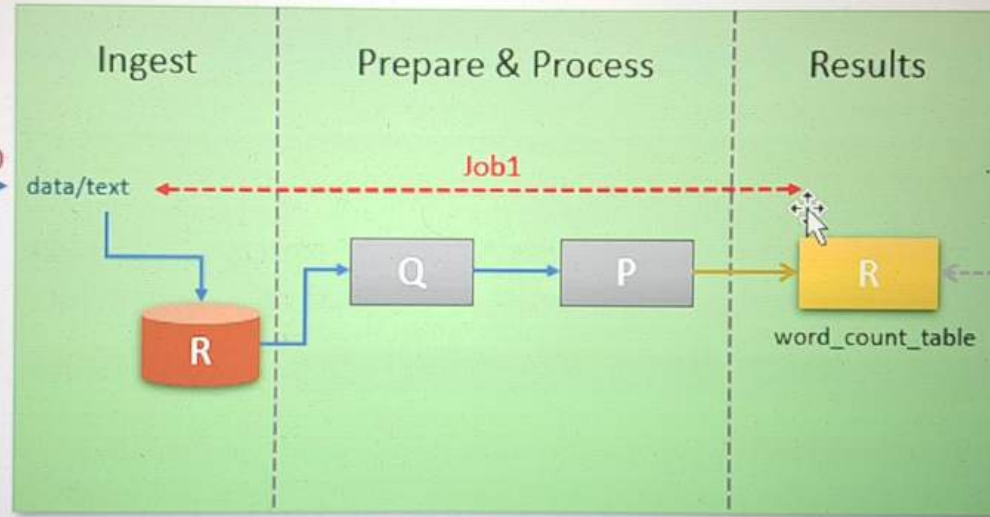


Udemy

Source System



Data Engineering Platform



Data Pipeline (Workflow)



What is the frequency?
15 minutes

Consumer Application

